# 06

## Video Recommendation System

Recommendation systems play a key role in video and music streaming services. For example, YouTube recommends videos a user may like, Netflix recommends movies a user may enjoy watching, and Spotify recommends music to users.

In this chapter, we design a video recommendation system similar to YouTube's [1]. The system recommends videos on the user's homepage based on their profile, previous interactions, etc.

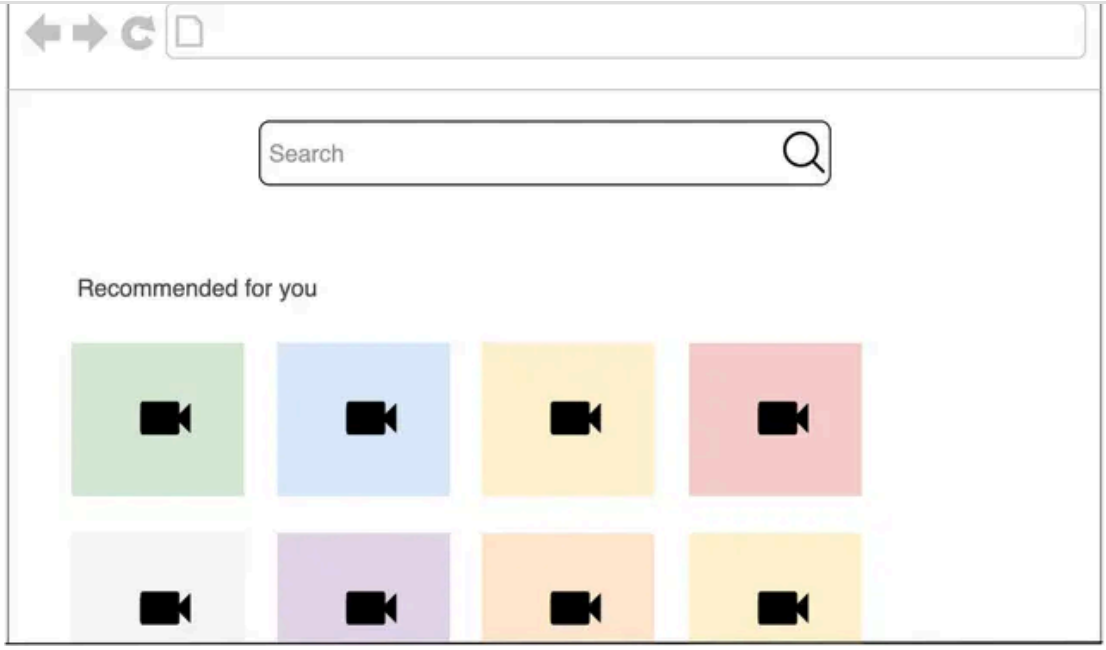



Figure 6.1: Homepage video recommendation

Recommendation systems are often very complex in design, and a good amount of engineering effort is required to develop an efficient and scalable system. Don't worry, though; no one expects you to build the perfect system in a 45 -minute interview. The interviewer is primarily interested in observing your thought process, communication skills, ability to design ML systems, and ability to discuss trade-offs.

## Clarifying Requirements

Here is a typical interaction between a candidate and an interviewer.

**Candidate:** Can I assume the business objective of building a video recommendation system is to increase user engagement?
**Interview**: That's correct.

**Candidate:** Does the system recommend similar videos to a video a user is watching right now? Or does it show a personalized list of videos on the user's homepage?
**Interviewer:** This is a homepage video recommendation system, which recommends personalized videos to users when they load the homepage.

**Candidate:** Since YouTube is a global service, can I assume users are located worldwide and videos are in different languages?
**Interviewer:** That's a fair assumption.

**Candidate:** Can I assume we can construct the dataset based on user interactions with video content?
**Interviewer:** Yes, that sounds good.

**Interviewer:** For the sake of simplicity, let's assume the playlist feature does not exist.

**Candidate:** How many videos are available on the platform?
**Interviewer:** We have about 10 billion videos.

**Candidate:** How fast should the system recommend videos to a user? Can I assume the recommendation should not take more than 200 milliseconds?
**Interviewer:** That sounds good.

Let's summarize the problem statement. We are asked to design a homepage video recommendation system. The business objective is to increase user engagement. Each time a user loads the homepage, the system recommends the most engaging videos. Users are located worldwide, and videos can be in different languages. There are approximately 10 billion videos on the platform, and recommendations should be served quickly.

## Frame the Problem as an ML Task

### Defining the ML objective

The business objective of the system is to increase user engagement. There are several options available for translating business objectives into well-defined ML objectives. We will examine some of them and discuss their trade-offs.

**Maximize the number of user clicks.** A video recommendation system can be designed to maximize user clicks. However, this objective has one major drawback. The model may recommend videos that are so-called "clickbait", meaning the title and thumbnail image look compelling, but the video's content may be boring, irrelevant, or even misleading. Clickbait videos reduce user satisfaction and engagement over time.

**Maximize the number of completed videos.** The system could also recommend videos users will likely watch to completion. A major problem with this objective is that the model may recommend shorter videos that are quicker to watch.

**Maximize total watch time.** This objective produces recommendations that users spend more time watching.

**Maximize the number of relevant videos.** This objective produces recommendations that are relevant to users. Engineers or product managers can define relevance based on some rules. Such rules can be based on implicit and explicit user reactions. For example, one definition could state a video is relevant if a user explicitly presses the "like" button or watches at least half of it. Once we define relevance, we can construct a dataset and train a model to predict the relevance score between a user and a video.

earlier.

### Specifying the system's input and output

As Figure 6.2 shows, a video recommendation system takes a user as input and outputs a ranked list of videos sorted by their relevance scores.

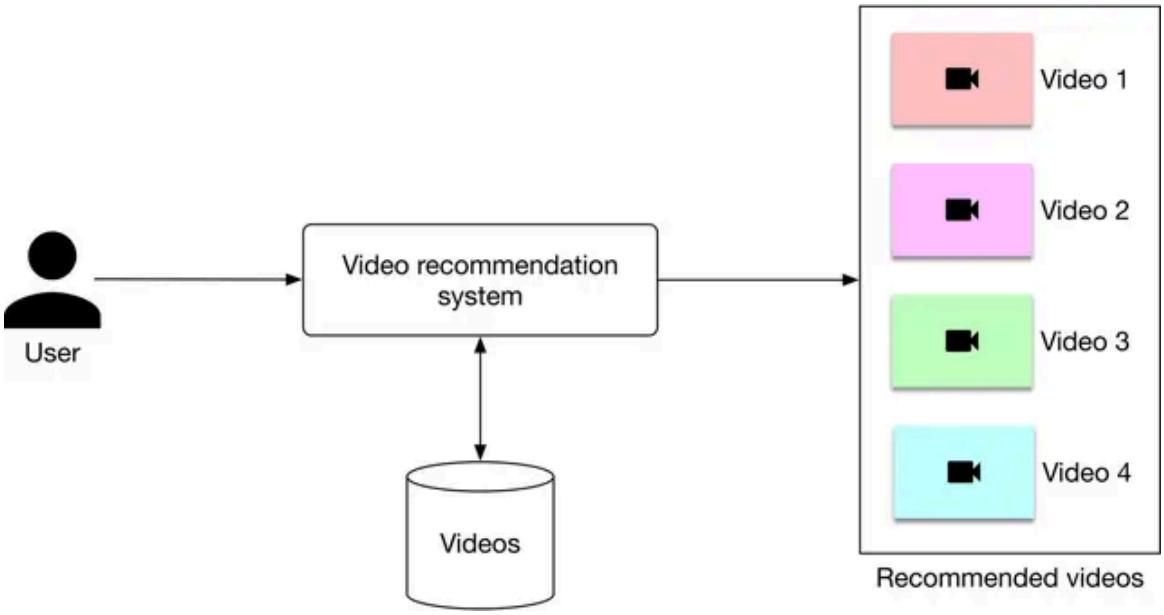



Figure 6.2: A video recommendation system's input-output

### Choosing the right ML category

In this section, we examine three common types of personalized recommendation systems.

- Content-based filtering
- Collaborative filtering
- Hybrid filtering

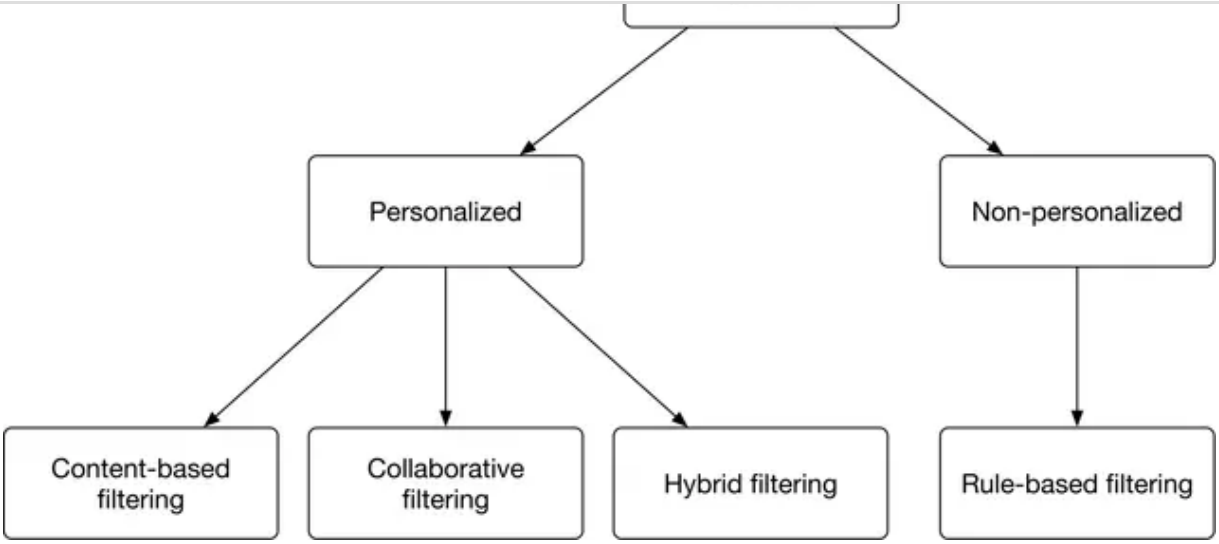



Figure 6.3: Common types of recommendation systems

Let's examine each type in more detail.

### Content-based filtering

This technique uses video features to recommend new videos similar to those a user found relevant in the past. For example, if a user previously engaged with many ski videos, this method will suggest more ski videos. Figure 6.4 shows an example.

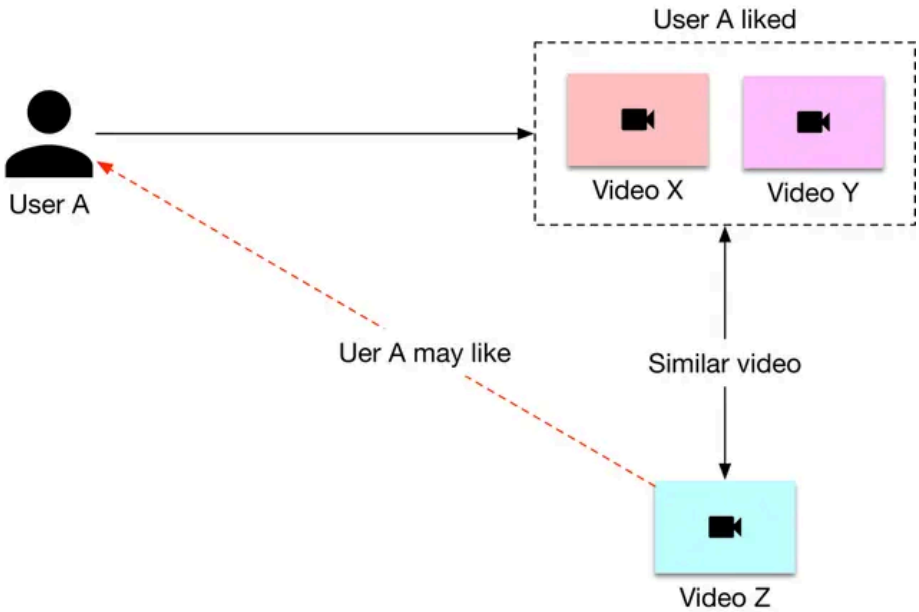



Figure 6.4: Content-based filtering

Here is an explanation of the diagram.

1. User A engaged with videos X and Y in the past

Content-based filtering has pros and cons.

**Pros:**

- **Ability to recommend new videos.** With this method, we don't need to wait for interaction data from users to build video profiles for new videos. The video profile depends entirely upon its features.
- **Ability to capture the unique interests of users.** This is because we recommend videos based on users' previous engagements.

**Cons:**

- **Difficult to discover a user's new interests.**
- The method requires **domain knowledge**. We often need to engineer video features manually.

### Collaborative filtering (CF)

CF uses user-user similarities (user-based CF) or video-video similarities (item-based CF) to recommend new videos. CF works with the intuitive idea that similar users are interested in similar videos. You can see a user-based CF example in Figure 6.5.

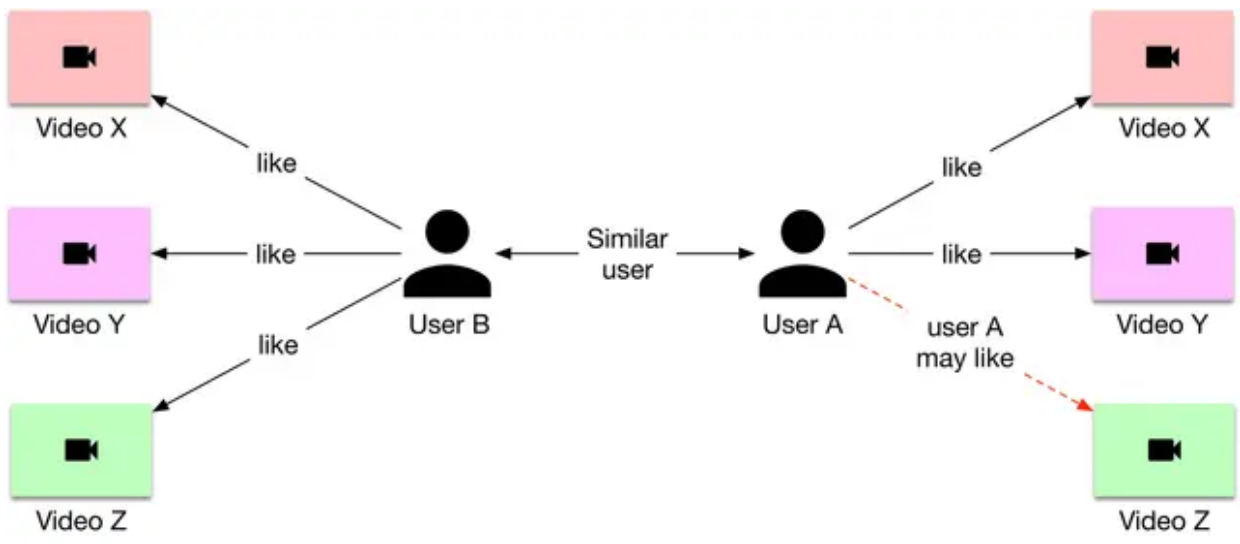



Figure 6.5: User-based collaborative filtering

Let's explain the diagram. The goal is to recommend a new video to user A.

1. Find a similar user to A based on their previous interactions; say user B
2. Find a video that user B engaged with but which user A has not seen yet; say video Z
3. Recommend video Z to user A

Let's see the pros and cons of CF filtering.

**Pros:**

- **No domain knowledge needed.** CF does not rely on video features, which means no domain knowledge is needed to engineer features from videos.
- **Easy to discover users' new areas of interest.** The system can recommend videos about new topics that other similar users engaged with in the past.
- **Efficient.** Models based on CF are usually faster and less compute-intensive than content-based filtering, as they do not rely on video features.

**Cons:**

- **Cold-start problem.** This refers to a situation when limited data is available for a new video or user, meaning the system cannot make accurate recommendations. CF suffers from a cold-start problem due to the lack of historical interaction data for new users or videos. This lack of interactions prevents CF from finding similar users or videos. We will discuss later in the serving section how our system handles the cold-start problem.
- **Cannot handle niche interests.** It's difficult for CF to handle users with specialized or niche interests. CF relies upon similar users to make recommendations, and it might be difficult to find similar users with niche interests.

| | Contentbased filtering | Collaborative filtering |
|---|---|---|
| Handle new videos | ✓ | ✗ |
| Discover new interest areas | ✗ | ✓ |
| No domain knowledge necessary | ✗ | ✓ |
| Efficiency | ✗ | ✓ |

Table 6.1: Comparison between content-based filtering and CF

A comparison of the two types of filtering is shown in Table 6.1. As you see, the two methods are complementary.

### Hybrid filtering

Hybrid filtering uses both CF and content-based filtering. As Figure 6.6 shows, hybrid filtering combines CF-based and content-based recommenders sequentially, or in parallel. In practice, companies usually use sequential hybrid filtering [2].

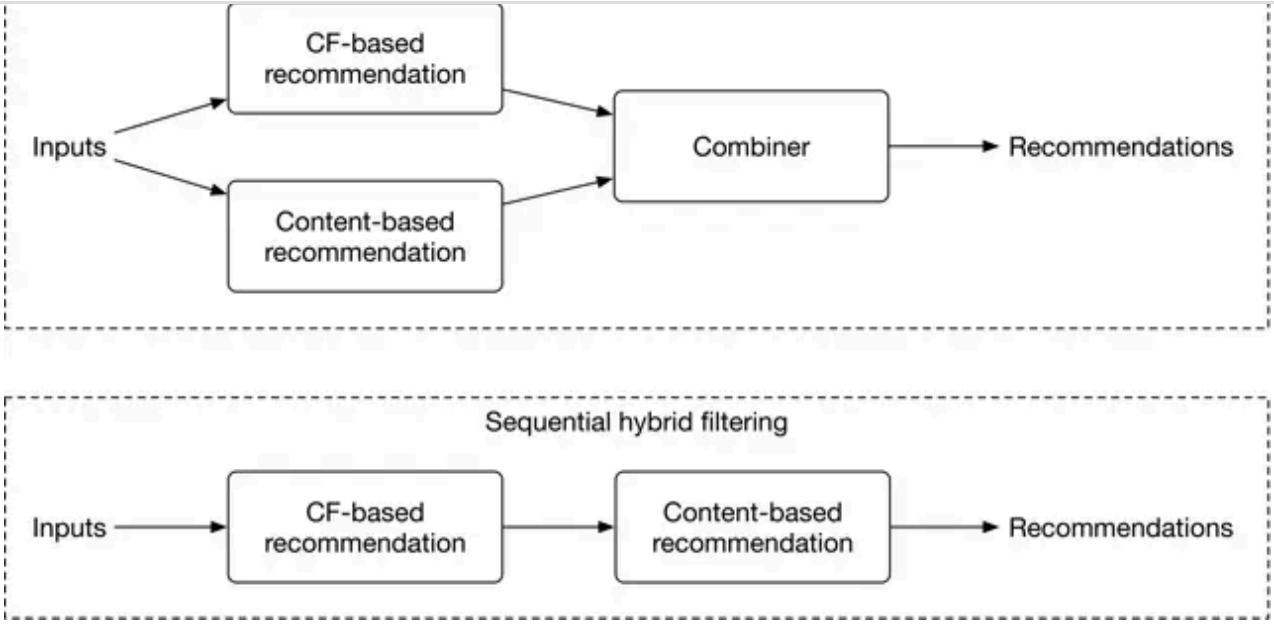



Figure 6.6: Hybrid filtering method

This approach leads to better recommendations because it uses two data sources: the user's historical interactions and video features. Video features allow the system to recommend relevant videos based on videos the user engaged with in the past, and CF-based filtering helps users to discover new areas of interest.

#### Which method should we choose?

Many companies use hybrid filtering to make better recommendations. For example, a paper published by Google [2] describes how YouTube employs a CF-based model as the first stage (candidate generator), followed by a content-based model as the second stage, to recommend videos. Due to the advantages of hybrid filtering, we choose this option.

## Data Preparation

### Data engineering

We have the following data available:

- Videos
- Users
- User-video interactions

#### Videos

Video data contains raw video files and their associated metadata, such as video ID, video length, video title, etc. Some of these attributes are provided explicitly by video uploaders, and others can be

| Video ID | Length | Manual tags | Manual title | Likes | Views | Language |
|---|---|---|---|---|---|---|
| 1 | 28 | Dog, Family | Our lovely dog playing! | 138 | 5300 | English |
| 2 | 300 | Car, Oil | How to change your car oil? | 5 | 250 | Spanish |
| 3 | 3600 | Ouli, Vlog | Honeymoon to Bali | 2200 | 255K | Arabic |

Table 6.2: Video metadata

### Users

The following simple schema represents user data.

| ID | Username | Age | Gender | City | Country | Language | Time zone |
|---|---|---|---|---|---|---|---|

Table 6.3: User data schema

### User-video interactions

The user-video interaction data consists of various user interactions with the videos, including likes, clicks, impressions, and past searches. Interactions are recorded along with other contextual information, such as location and timestamp. The following table shows how user-video interactions are stored.

| User ID | Video ID | Interaction type | Interaction value | Location (lat, long) | Timestamp |
|---|---|---|---|---|---|
| 4 | 18 | Like | - | 38.8951<br>-77.0364 | 1658451361 |
| 2 | 18 | Impression | 8 seconds | 38.8951<br>-77.0364 | 1658451841 |
| 2 | 6 | Watch | 46 minutes | 41.9241<br>-89.0389 | 1658822820 |
| 6 | 9 | Click | - | 22.7531<br>47.9642 | 1658832118 |
| 9 | - | Search | Basics of clustering | 22.7531<br>47.9642 | 1659259402 |

| | | | | | |
|---|---|---|---|---|---|
| 8 | 6 | Comment | Amazing video. Thanks | 37.5189 122.6405 | 1659244197 |

Table 6.4: User-video interaction data

### Feature engineering

The ML system is required to predict videos that are relevant to users. Let's engineer features to help the system make informed predictions.

#### Video features

Some important video features include:

- Video ID
- Duration
- Language
- Titles and tags

#### Video ID

The IDs are categorical data. To represent them by numerical vectors, we use an embedding layer, and the embedding layer is learned during model training.

#### Duration

This defines approximately how long the video lasts from start to finish. This information is important since some users may prefer shorter videos, while others prefer longer videos.

#### Language

The language used in a video is an important feature. This is because users naturally prefer particular languages. Since language is a categorical variable and takes on a finite set of discrete values, we use an embedding layer to represent it.

#### Titles and tags

Titles and tags are used to describe a video. They are either provided manually by the uploader or are implicitly predicted by standalone ML models. The titles and tags of a video are valuable predictors. For example, a video titled "how to make pizza" indicates the video is related to pizza and cooking.

For the title, we map it into a feature vector using a context-aware word embedding model, such as a pre-trained BERT [4].

Figure 6.7 shows an overview of video feature preparation.

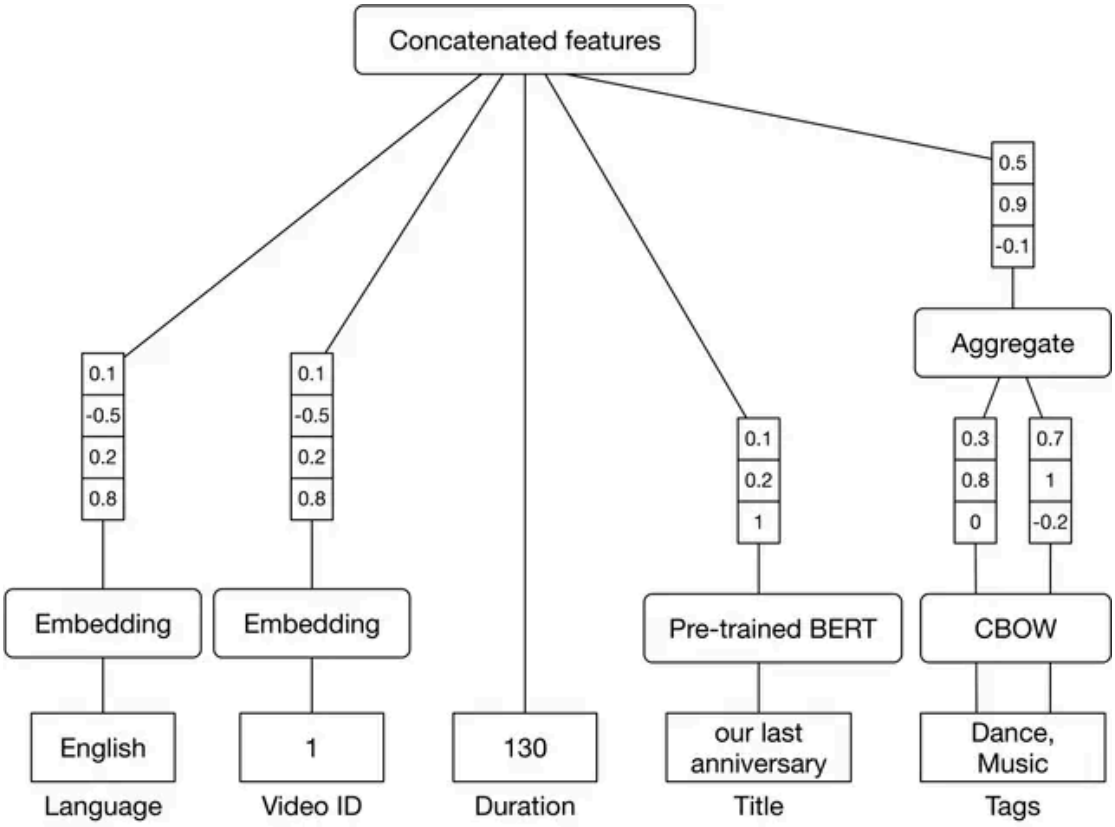



Figure 6.7: Video feature preparation

### User features

We categorize user features into the following buckets:

- User demographics
- Contextual information
- User historical interactions

### User demographics

An overview of user demographic features is shown in Figure 6.8.

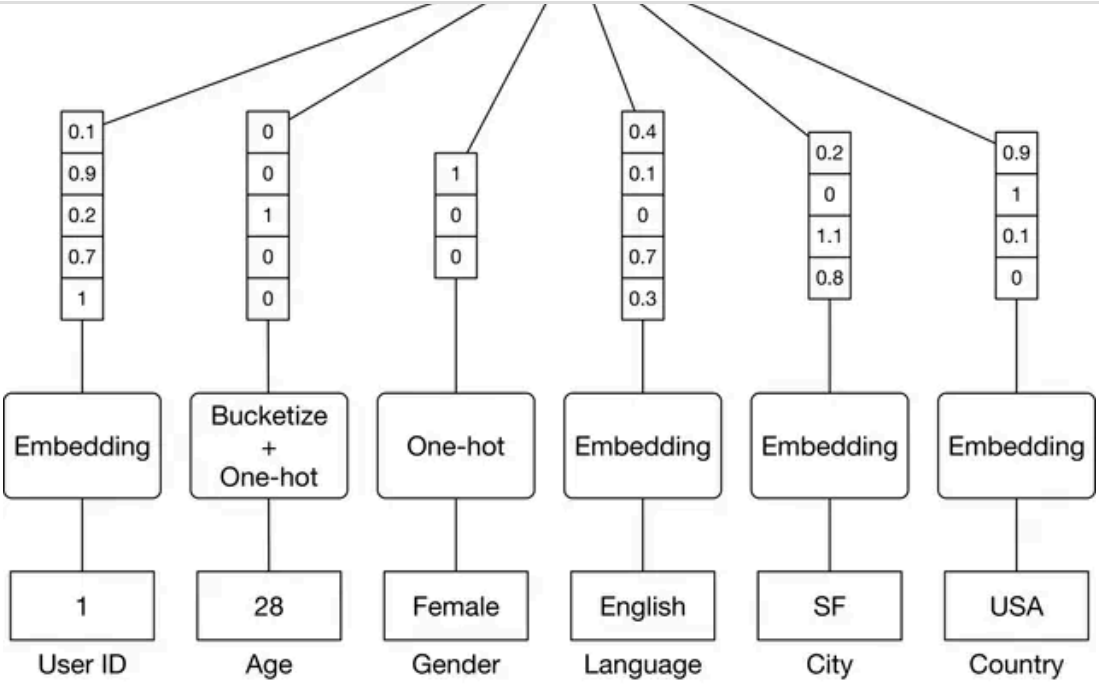



Figure 6.8: Features based on a user's demographics

### Contextual information

Here are a few important features for capturing contextual information:

- **Time of day.** A user may watch different videos at different times of day. For example, a software engineer may watch more educational videos during the evening.
- **Device.** On mobile devices, users may prefer shorter videos.
- **Day of the week.** Depending on the day of the week, users may have different preferences for videos.

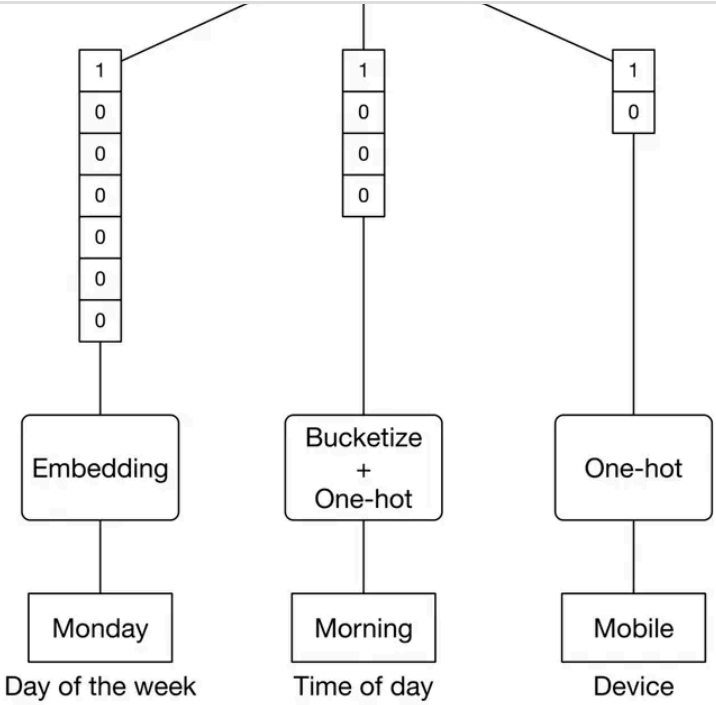



Figure 6.9: Features related to contextual information

### User historical interactions

User historical interactions play an important role in understanding user interests. A few features related to historical interactions are:

- Search history
- Liked videos
- Watched videos and impressions

**Search history**

**Why is it important?**

Previous searches indicate what the user looked for in the past, and past behavior is often an indicator of future behavior.

**How to prepare it?**

Use a pre-trained word embedding model, such as BERT, to map each search query into an embedding vector. Note that a user's search history is a variable-sized list of textual queries. To create a fixed-size feature vector summarizing all the search queries, we average the query embeddings.

**Liked videos**

**Why is it important?**

**How to prepare it?**

Video IDs are mapped into embedding vectors using the embedding layer. Similarly to search history, we average liked embeddings to get a fixed-size vector of liked videos.

**Watched videos and impressions**

The feature engineering process for “watched videos" and “impressions" is very similar to what we did for liked videos. So, we won’t repeat it.

Figure 6.10 summarizes features related to user-video interactions.

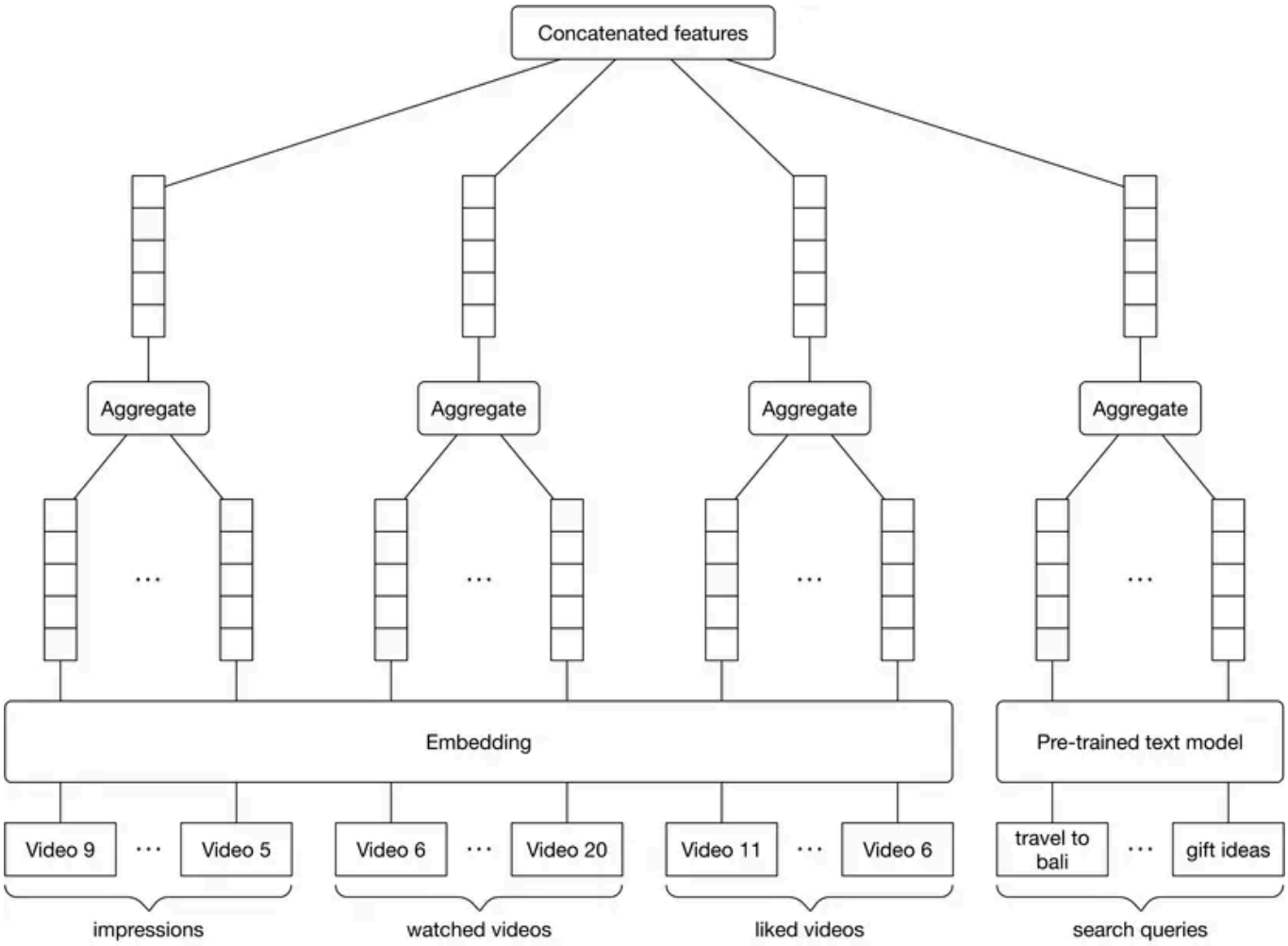



Figure 6.10: Features related to user-video interactions

## Model Development

In this section, we examine two embedding-based models that are typically employed in CF-based or content-based recommenders:

- Matrix factorization

### Matrix factorization

To understand the matrix factorization model, it is important to know what a feedback matrix is.

#### Feedback matrix

Also called a utility matrix, this is a matrix that represents users' opinions about videos. Figure 6.11 shows a binary user-video feedback matrix where each row represents a user, and each column represents a video. The entries in the matrix specify the user's opinion to 1 as "observed" or "positive."

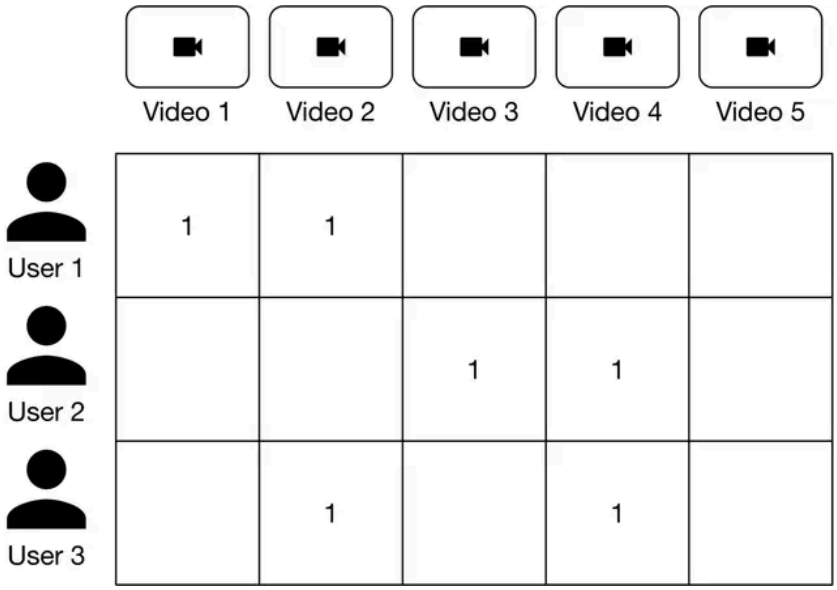



Figure 6.11: User-video feedback matrix

How can we determine whether a user finds a recommended video relevant? We have three options:

- Explicit feedback
- Implicit feedback
- Combination of explicit and implicit feedback

**Explicit feedback.** A feedback matrix is built based on interactions that explicitly indicate a user's opinion about a video, such as likes and shares. Explicit feedback reflects a user's opinion accurately as users explicitly expressed their interest in a video. This option, however, has one major drawback: the matrix is sparse since only a small fraction of users provide explicit feedback. Sparsity makes ML models difficult to train.

**Implicit feedback.** This option uses interactions that implicitly indicate a user's opinion about a video, such as "clicks" or "watch time". With implicit feedback, more data points are available, resulting in a better model after training. Its main disadvantage is that it does not directly reflect users' opinions and might be noisy.

**What is the best option for building our feedback matrix?**

Since the model needs to learn the values of the feedback matrix, it's important to build the matrix that aligns well with the ML objective we chose earlier.

In our case, the ML objective is to maximize relevancy, where relevancy is defined as the combination of explicit and implicit feedback. As such, the final option of combining explicit and implicit feedback is the best choice.

### Matrix factorization model

Matrix factorization is a simple embedding model. The algorithm decomposes the user-video feedback matrix into the product of two lower-dimensional matrices. One lower-dimensional matrix represents user embeddings, and the other represents video embeddings. In other words, the model learns to map each user into an embedding vector and each video into an embedding vector, such that their distance represents their relevance. Figure 6.12 shows how a feedback matrix is decomposed into user and video embeddings.

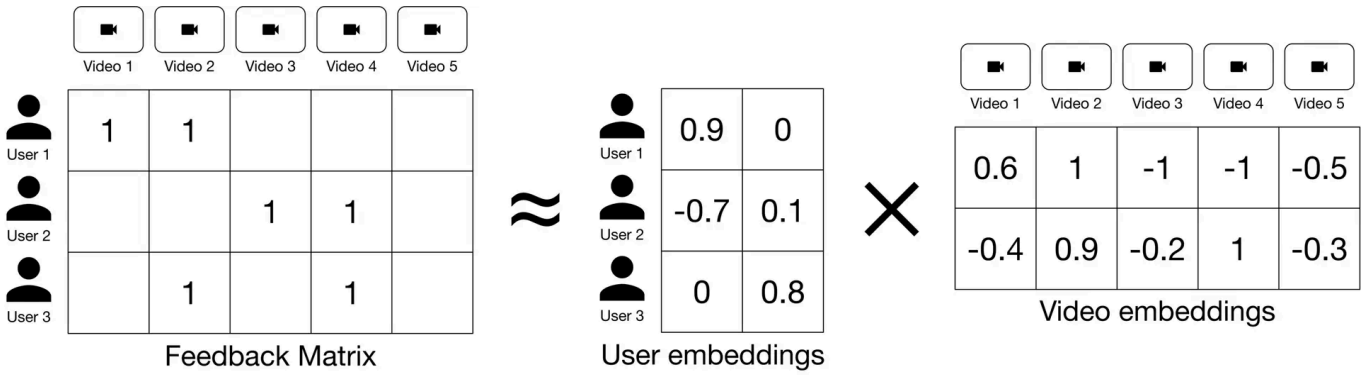



Figure 6.12: Decompose the feedback matrix into two matrices

### Matrix factorization training

As part of training, we aim to produce user and video embedding matrices so that their product is a good approximation of the feedback matrix (Figure 6.13.)

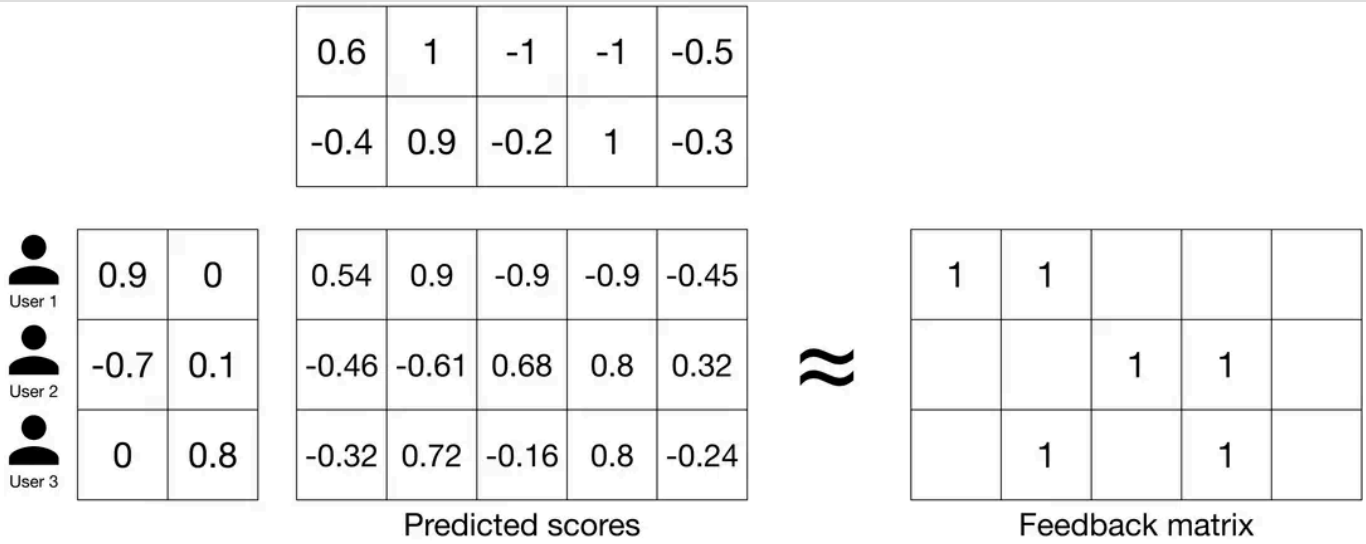


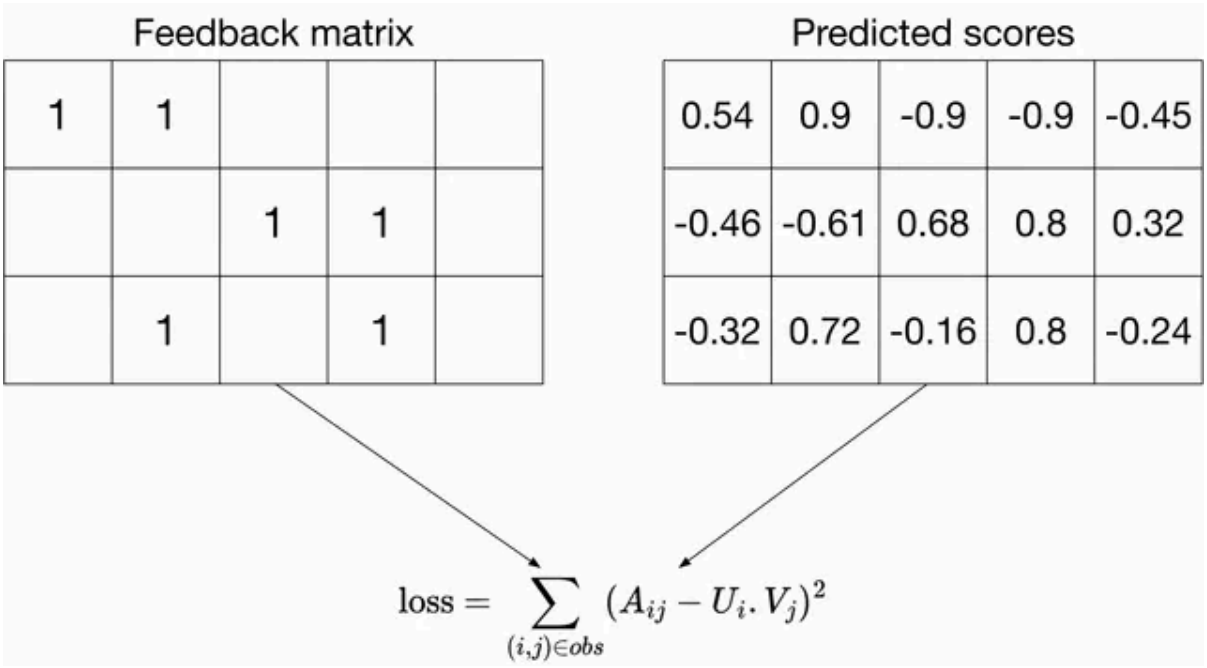

Figure 6.13: The product of embeddings should approximate the feedback matrix

To learn these embeddings, matrix factorization first randomly initializes two embedding matrices, then iteratively optimizes the embeddings to decrease the loss between the "Predicted scores matrix" and the "Feedback matrix". Loss function selection is an important consideration. Let's explore a few options:

- Squared distance over observed ⟨ user, video ⟩ pairs
- A weighted combination of squared distance over observed pairs and unobserved pairs

**Squared distance over observed ⟨ user, video ⟩ pairs**

This loss function measures the sum of the squared distances over all pairs of observed (non-zero values) entries in the feedback matrix. This is shown in Figure 6.14.



$A_{ij}$ refers to the entry with row $i$ and column $j$ in the feedback matrix, $U_i$ is the embedding of user $i$, $V_j$ is the embedding of video $j$, and the summation is over the observed pairs only.

Only summing over observed pairs leads to poor embeddings because the loss function doesn't penalize the model for bad predictions on unobserved pairs. For example, embedding matrices of all ones would have a zero loss on the training data. However, those embeddings may not work well for unseen ⟨ user, video ⟩ pairs.

**Squared distance over both observed and unobserved ⟨ user, video ⟩ pairs** This loss function treats unobserved pairs as negative data points and assigns a zero to them in the feedback matrix. As Figure 6.15 shows, the loss computes the sum of the squared distances over all entries in the feedback matrix.

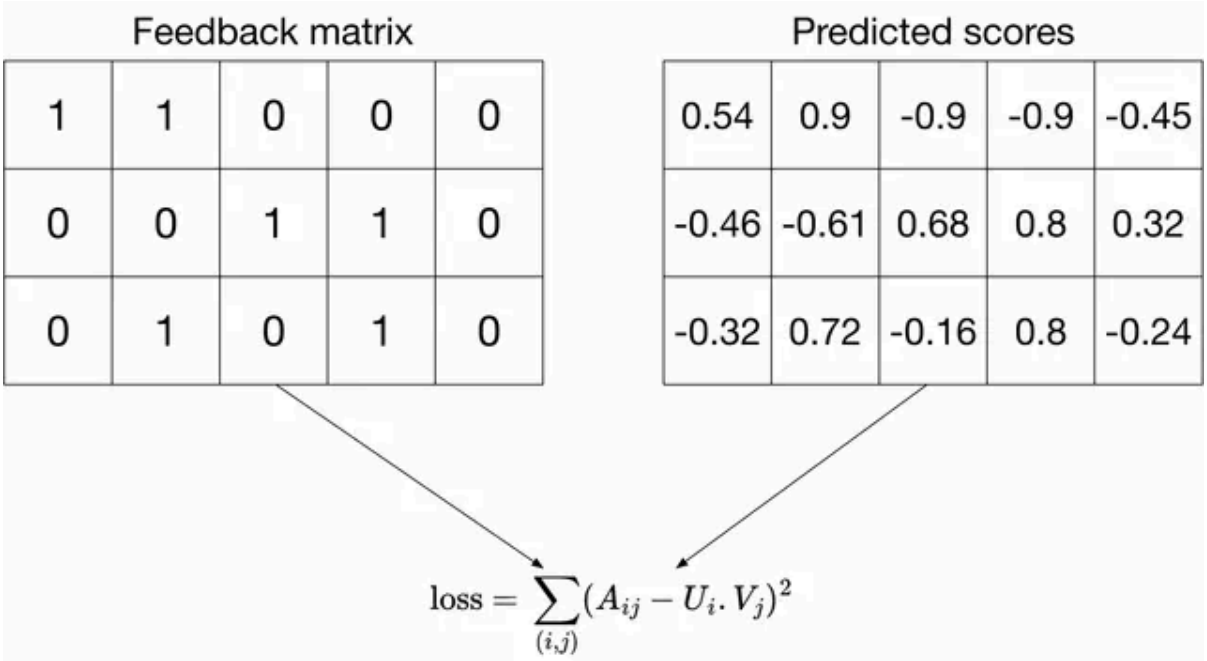



Figure 6.15: Squared distance over all ⟨ user, video ⟩ pairs

This loss function addresses the previous issue by penalizing bad predictions for unobserved entries. However, this loss has a major drawback. The feedback matrix is usually sparse (lots of unobserved pairs), so unobserved pairs dominate observed pairs during training. This results in predictions that are mostly close to zero. This is not desirable and leads to poor generalization performance on unseen ⟨ user, video ⟩ pairs.

**A weighted combination of squared distance over observed and unobserved pairs**
To overcome the drawbacks of the loss functions described earlier, we opt for weighted combinations of both.

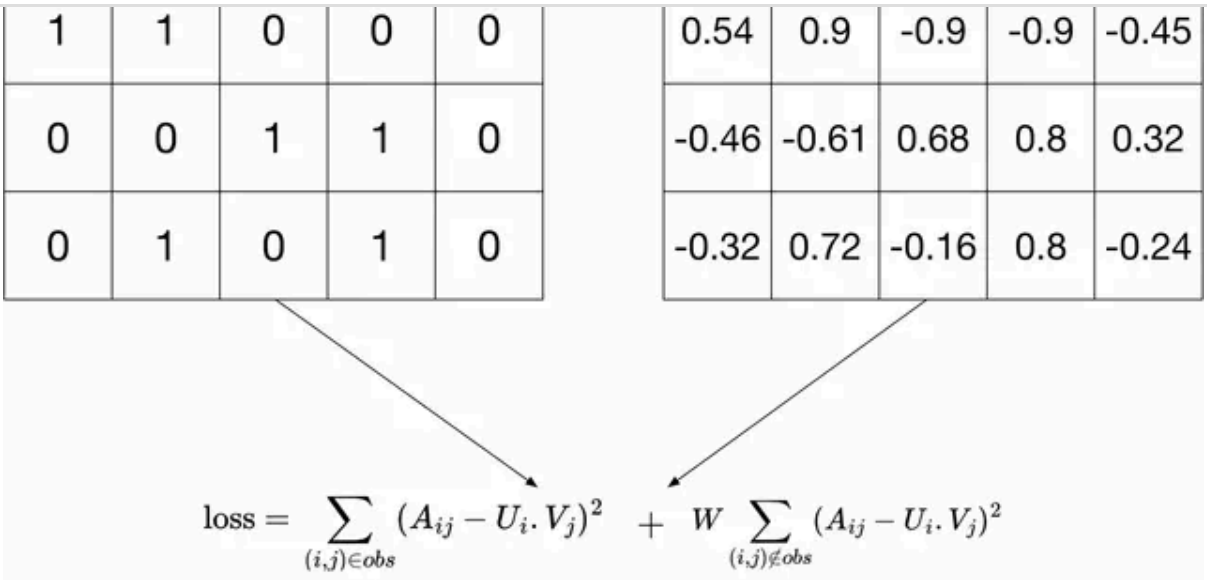



Figure 6.16: Combined loss

The first summation in the loss formula calculates the loss on the observed pairs, and the second summation calculates the loss on unobserved pairs. $W$ is a hyperparameter that weighs the two summations. It ensures one does not dominate the other in the training phase. This loss function with a properly tuned $W$ works well in practice [5]. We choose this loss function for the system.

### Matrix factorization optimization

To train an ML model, an optimization algorithm is required. Two commonly used optimization algorithms in matrix factorization are:

- Stochastic Gradient Descent (SGD): This optimization algorithm is used to minimize losses [6].
- Weighted Alternating Least Squares (WALS): This optimization algorithm is specific to matrix factorization. The process in WALS is:
  1. Fix one embedding matrix (U), and optimize the other embedding (V)
  2. Fix the other embedding matrix (V), and optimize the embedding matrix (U)
  3. Repeat.

  WALS usually converges faster and is parallelizable. To learn more about WALS, read [7]. Here, we use WALS because it converges faster.

### Matrix factorization inference

To predict the relevance between an arbitrary user and a candidate video, we calculate the similarity between their embeddings using a similarity measure, such as a dot product. For example, as shown in Figure 6.17, the relevance score between user 2 and video 5 is 0.32.

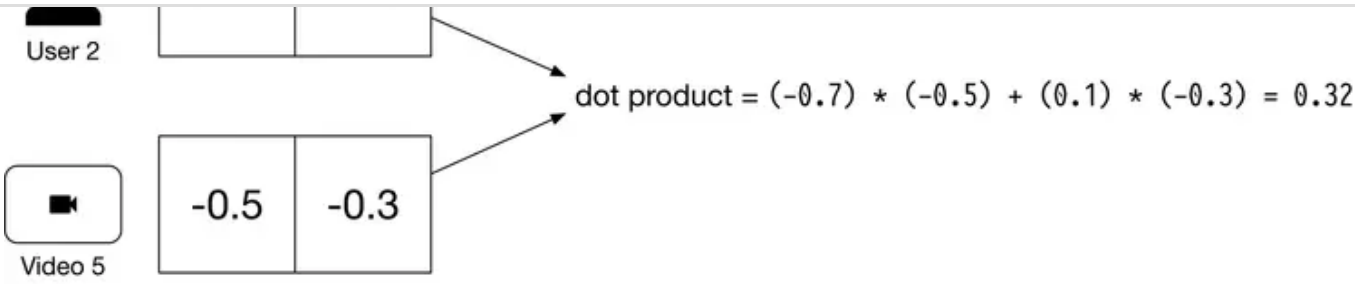



Figure 6.17: Relevance score for ⟨ user 2, video 5 ⟩ pair

Figure 6.18 shows the predicted scores for all the ⟨ user, video ⟩ pairs. The system returns recommended videos based on relevance scores.

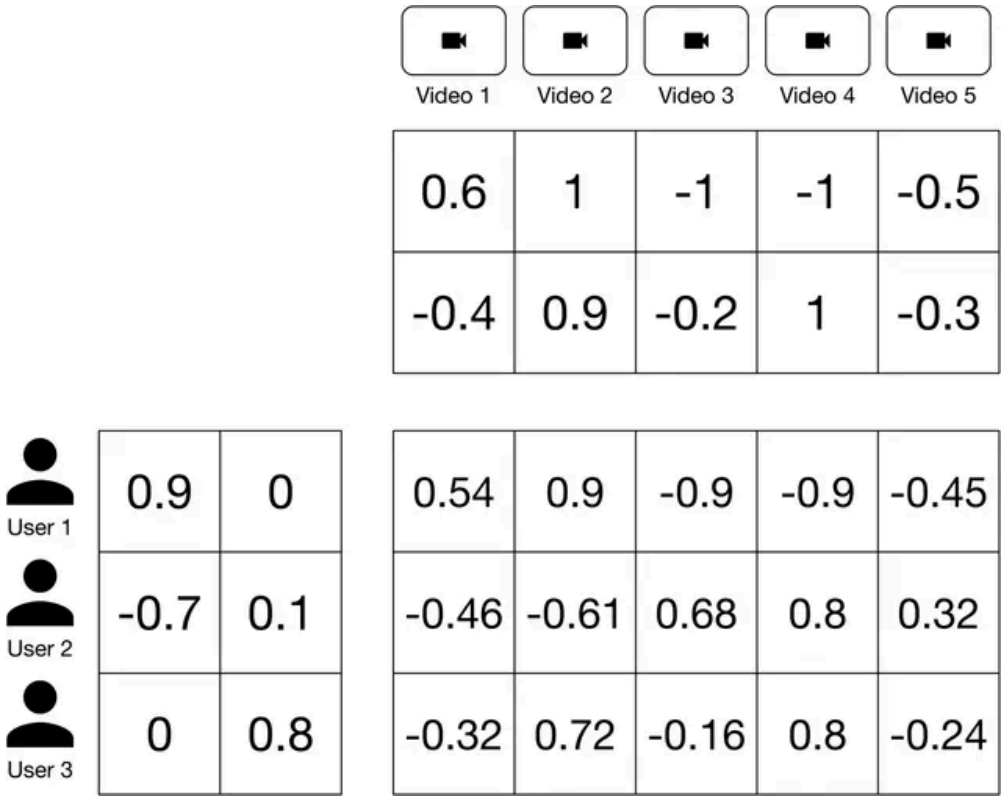



Figure 6.18: Predicted pairwise relevance scores

| **Reminder** |
| --- |
| Since matrix factorization uses user-video interactions only, it is commonly used in collaborative filtering. |

Before wrapping up matrix factorization, let's discuss the pros and cons of this model.

**Pros:**

- Training speed: Matrix factorization is efficient during the training phase. This is because there are only two embedding matrices to learn.

query time.

**Cons:**

- Matrix factorization only relies on user-video interactions. It does not use other features, such as the user's age or language. This limits the predictive capability of the model because features like language are useful to improve the quality of recommendations.
- Handling new users is difficult. For new users, there are not enough interactions for the model to produce meaningful embeddings. Therefore, matrix factorization cannot determine whether a video is relevant to a user by computing the dot product between their embeddings.

Let's see how two-tower neural networks address the shortcomings of matrix factorization.

### Two-tower neural network

A two-tower neural network comprises two encoder towers: the user tower and the video tower. The user encoder takes user features as input and maps them to an embedding vector (user embedding). The video encoder takes video features as input and maps them into an embedding vector (video embedding). The distance between their embeddings in the shared embedding space represents their relevance.

Figure 6.19 shows the two-tower architecture. In contrast to matrix factorization, twotower architectures are flexible enough to incorporate all kinds of features to better capture the user's specific interests.

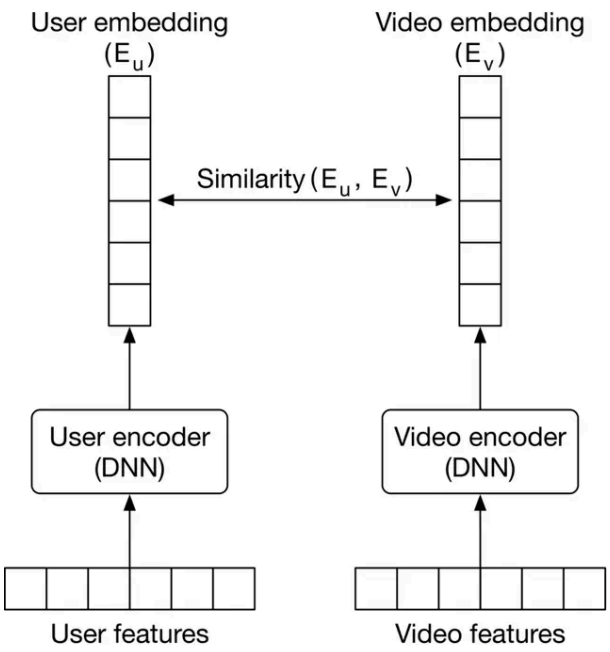



Figure 6.19: Two-tower neural network

as positive or negative based on the user's feedback. For example, we label a pair as "positive" if the user explicitly liked the video, or watched at least half of it.

To construct negative data points, we can either choose random videos which are not relevant or choose those the user explicitly disliked by pressing the dislike button. Figure 6.20 shows an example of the constructed data points.

| # | User-related features | Video-related features | Label |
|---|---|---|---|
| 1 | 0 0 1 0.7 -0.6 0 0 | 0 1 0 0.9 0.9 1 | 1 (positive) |
| 2 | 0 1 1 0.2 0.1 1 0 | 0 1 0 -0.1 0.3 1 | 0 (negative) |

Figure 6.20: Two constructed data points

Note, users usually only find a small fraction of videos relevant. While constructing training data, this leads to an imbalanced dataset where there are many more negative than positive pairs. Training a model on an imbalanced dataset is problematic. We can use the techniques described in Chapter 1 Introduction and Overview, to address the data imbalance issue.

### Choosing the loss function

Since the two-tower neural network is trained to predict binary labels, the problem can be categorized as a classification task. We use a typical classification loss function, such as cross-entropy, to optimize the encoders during training. This process is shown in Figure 6.21

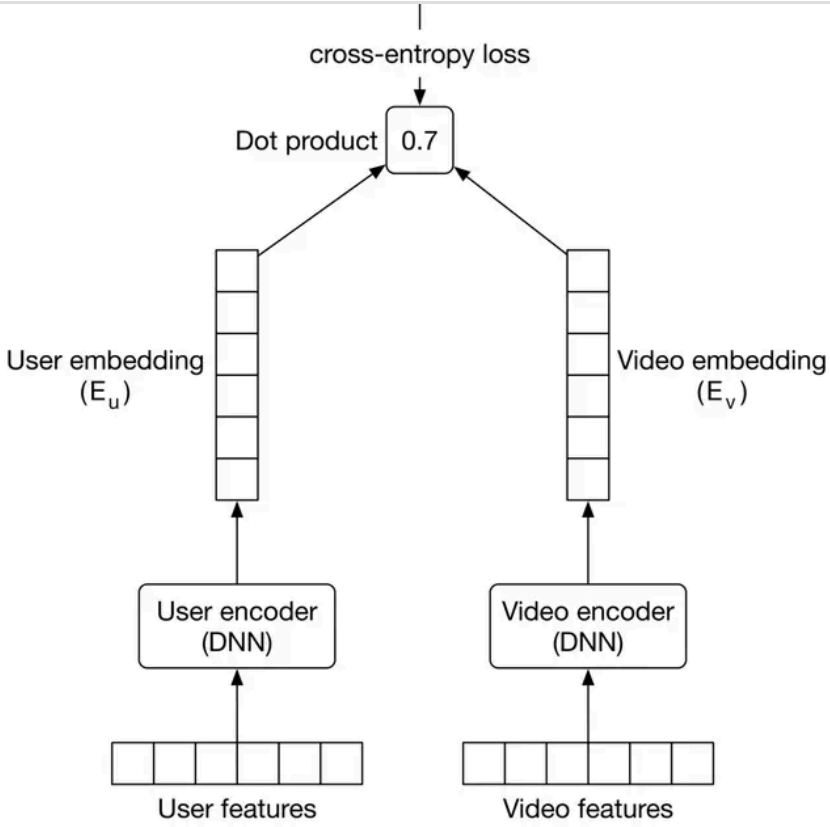



Figure 6.21: Two-tower neural network training workflow

### Two-tower neural network inference

At inference time, the system uses the embeddings to find the most relevant videos for a given user. This is a classic "nearest neighbor" problem. We use approximate nearest neighbor methods to find the top k most similar video embeddings efficiently.

Two-tower neural networks are used for both content-based filtering and collaborative filtering. When a two-tower architecture is used for collaborative filtering, as shown in Figure 6.22, the video encoder is nothing but an embedding layer that converts the video ID into an embedding vector. This way, the model doesn't rely on other video features.

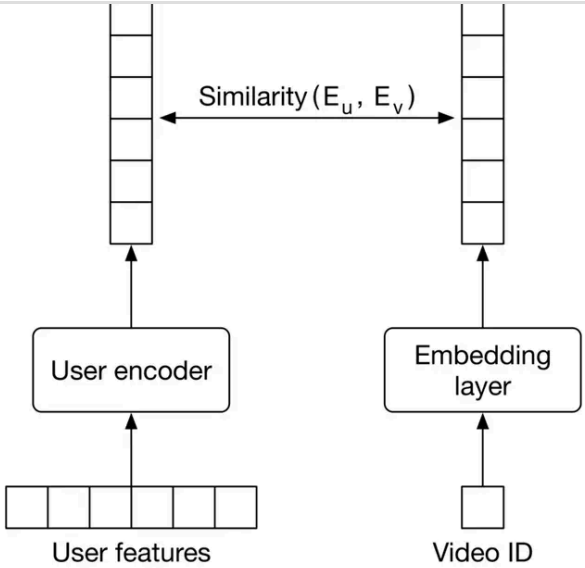



Figure 6.22: Two-tower neural network used for collaborative filtering

Let's see the pros and cons of a two-tower neural network model.

**Pros:**

- **Utilizes user features.** The model accepts user features, such as age and gender, as input. These predictive features help the model make better recommendations.
- **Handles new users.** The model easily handles new users as it relies on user features (e.g., age, gender, etc.).

**Cons:**

- **Slower serving.** The model needs to compute the user embedding at query time. This makes the model slower to serve requests. In addition, if we use the model for content-based filtering, the model needs to transform video features into video embedding, which increases the inference time.
- **Training is more expensive.** Two-tower neural networks have more learning parameters than matrix factorization. Therefore, the training is more compute-intensive.

### Matrix factorization vs. two-tower neural network

Table 6.5 summarizes the differences between matrix factorization and two-tower neural network architecture.

| | **Matrix factorization** | **Two-tower neural network** |
|---|---|---|
| Training cost | ✓ More efficient to train | ✗ More costly to train |

| | | |
|---|---|---|
| Inference speed | ✓ Faster as embeddings are static and can be precomputed | ✗ User features should be transformed into embeddings at query time |
| Cold-start problem | ✗ Cannot handle new users easily | ✓ Handles new users as it relies on user features |
| Quality of recommendations | ✗ Not ideal since the model does not use user/video features | ✓ Better recommendations since it relies on more features |

Table 6.5: Matrix factorization vs. two-tower neural networks

## Evaluation

The system's performance can be evaluated with offline and online metrics.

### Offline metrics

We evaluate the following offline metrics commonly used in recommendation systems.

**Precision@k.** This metric measures the proportion of relevant videos among the top $k$ recommended videos. Multiple $k$ values (e.g., $1, 5, 10$ ) can be used.

**mAP.** This metric measures the ranking quality of recommended videos. It is a good fit because the relevance scores are binary in our system.

**Diversity.** This metric measures how dissimilar recommended videos are to each other. This metric is important to track, as users are more interested in diversified videos. To measure diversity, we calculate the average pairwise similarity (e.g., cosine similarity or dot product) between videos in the list. A low average pairwise similarity score indicates the list is diverse.

Note that using diversity as the sole measure of quality can result in misleading interpretations. For example, if the recommended videos are diverse but irrelevant to the user, they may not find the recommendations helpful. Therefore, we should use diversity with other offline metrics to ensure both relevance and diversity.

### Online metrics

In practice, companies track many metrics during online evaluation. Let's examine some of the most important ones:

- Click-through rate (CTR)
- The number of completed videos
- Total watch time

CTR: The ratio between clicked videos and the total number of recommended videos. The formula is:

$$CTR = \frac{\text{number of clicked videos}}{\text{total number of recommended videos}}$$

CTR is an insightful metric to track user engagement, but the drawback of CTR is that we cannot capture or measure clickbait videos.

**The number of completed videos.** The total number of recommended videos that users watch until the end. By tracking this metric, we can understand how often the system recommends videos that users watch.

**Total watch time.** The total time users spent watching the recommended videos. When recommendations interest users, they spend more time watching videos, overall.

**Explicit user feedback.** The total number of videos that users explicitly liked or disliked. The metric accurately reflects users' opinions of recommended videos.

## Serving

At serving time, the system recommends the most relevant videos to a given user by narrowing the selection down from billions of videos. In this section, we will propose a prediction pipeline that's both efficient and accurate at serving requests.

Given we have billions of videos available, the serving speed would be slow if we choose a heavy model which takes lots of features as input. On the other hand, if we choose a lightweight model, it may not produce high-quality recommendations. So, what to do? A natural decision is to use more than one model in a multi-stage design. For example, in a two-stage design, a lightweight model quickly narrows down the videos during the first stage, called candidate generation. The second stage uses a heavier model that accurately scores and ranks the videos, called scoring. Figure 6.23 shows how candidate generation and scoring work together to produce relevant videos.

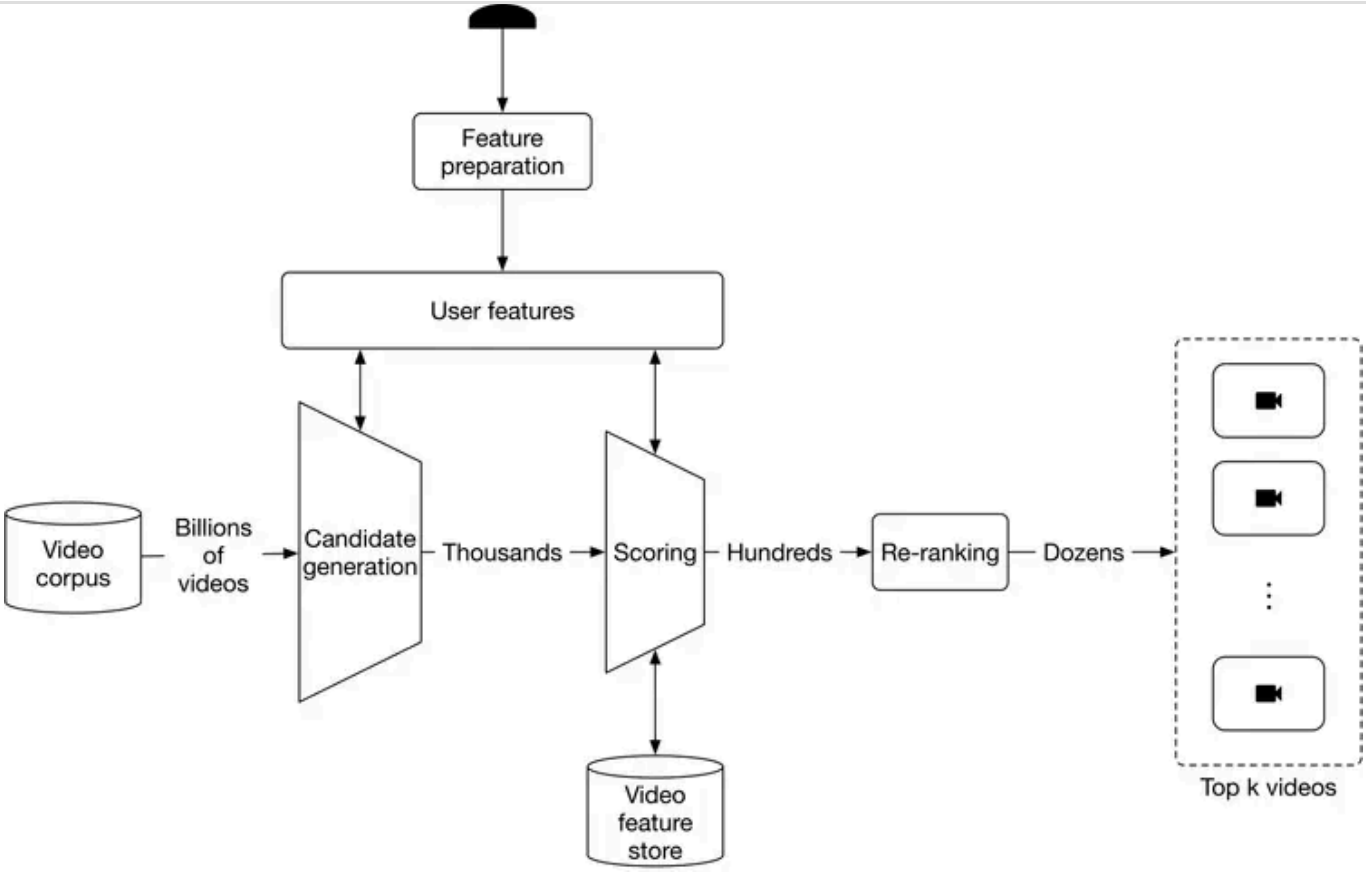



Figure 6.23: Prediction pipeline

Let's take a closer look at the components of the prediction pipeline.

- Candidate generation
- Scoring
- Re-ranking

### Candidate generation

The goal of candidate generation is to narrow down the videos from potentially billions, to thousands. We prioritize efficiency over accuracy at this stage and are not concerned about false positives.

To keep candidate generation fast, we choose a model which doesn't rely on video features. In addition, this model should be able to handle new users. A two-tower neural network is a good fit for this stage.

Figure 6.24 shows the candidate generation workflow. The candidate generation obtains a user's embedding from the user encoder. Once the computation is complete, it retrieves the most similar videos from the approximate nearest neighbor service. These videos are ranked based on similarity in the embedding space and are returned as the output.

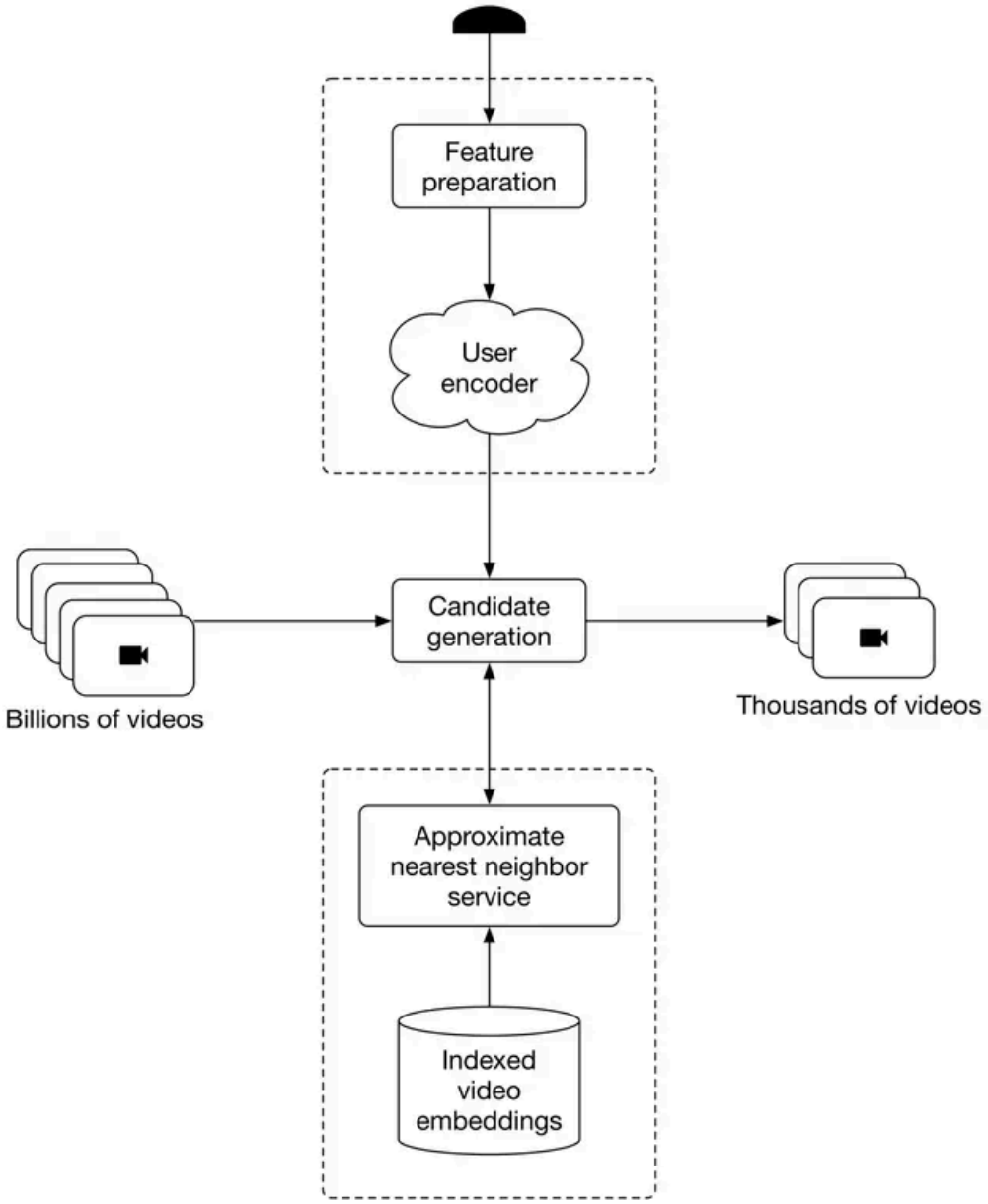



Figure 6.24: Candidate generation workflow

In practice, companies may choose to use more than one candidate generation because it could improve the performance of the recommendation. Let's take a look at why.

Users may be interested in videos for many reasons. For example, a user may choose to watch a video because it's popular, trending, or relevant to their location. To include those videos in the recommendations, it is common to use more than one candidate generation, as shown in Figure 6.25.

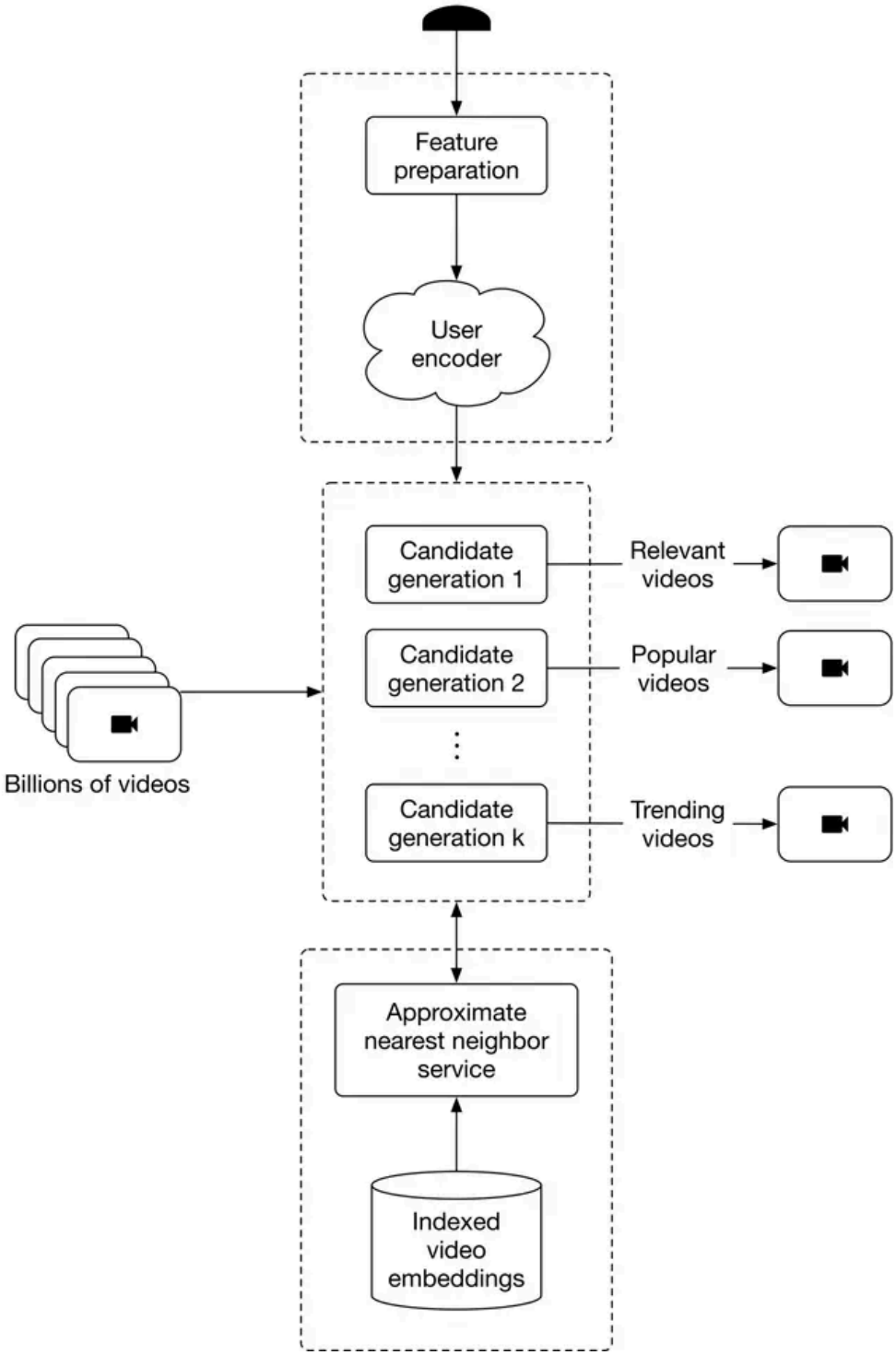



Figure 6.25: Use k candidate generations to diversify recommended videos

As soon as we have narrowed down potential videos from billions to thousands, we can use a scoring component to rank these videos before they are displayed.

### Scoring

Also known as ranking, scoring takes the user and candidate videos as input, scores each video, and outputs a ranked list of videos.

choice for this stage. Since there are only a handful of videos to rank in the scoring stage, we can employ a heavier model with more parameters. Figure 6.26 shows an overview of the scoring component.

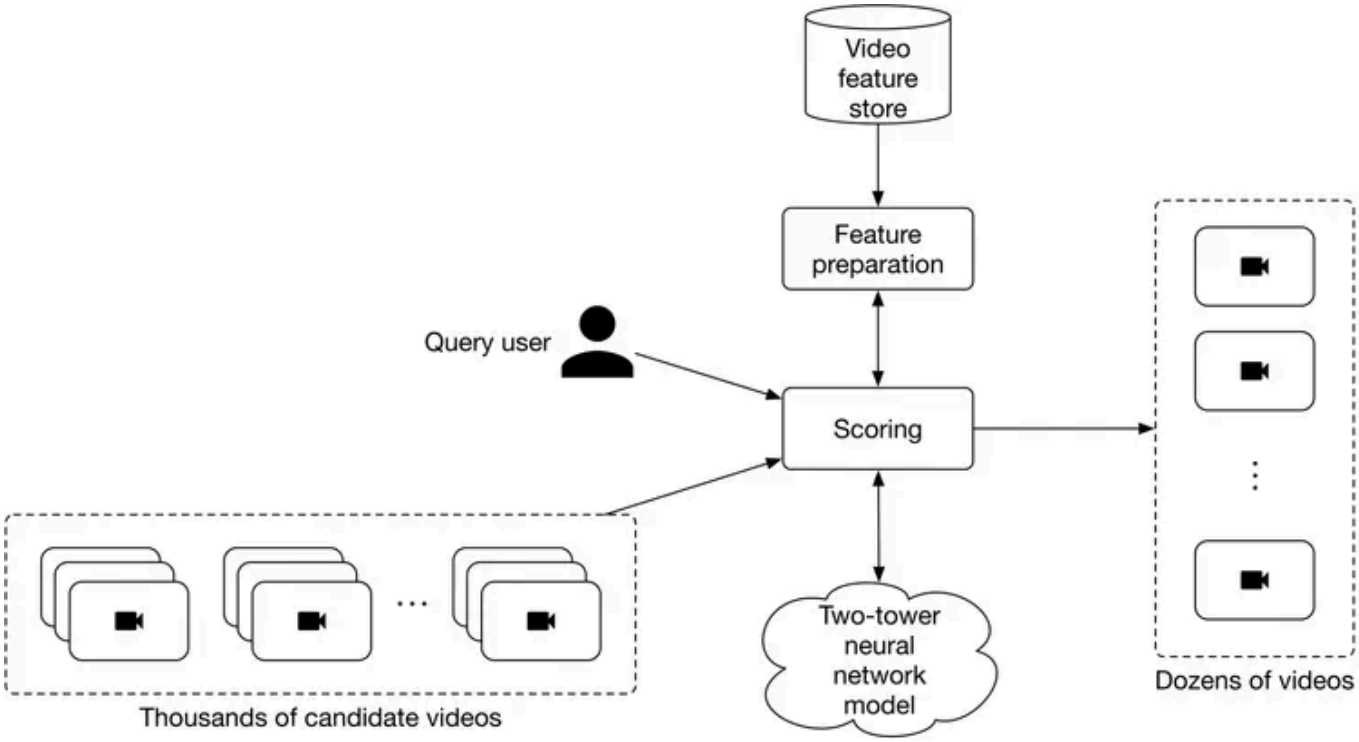



Figure 6.26: Overview of the scoring component

### Re-ranking

This component re-ranks the videos by adding additional criteria or constraints. For example, we may use standalone ML models to determine if a video is clickbait. Here are a few important things to consider when building the re-ranking component:

- Region-restricted videos
- Video freshness
- Videos spreading misinformation
- Duplicate or near-duplicate videos
- Fairness and bias

### Challenges of video recommendation systems

Before wrapping up this chapter, let's see how our design addresses typical challenges in video recommendation systems.

#### Serving speed

stage design.

Specifically, we use a lightweight model in the first stage to quickly narrow down candidate videos from billions to thousands. YouTube uses a similar approach [2] , and Instagram adopts a multi-stage design [8].

### Precision

To ensure precision, we employ a scoring component that ranks videos using a powerful model, which relies on more features, including video features. Using a more powerful model doesn't affect serving speed because only a small subset of videos is selected after the candidate generation phase.

### Diversity

Most users prefer to see a diverse selection of videos in their recommendations. To ensure our system produces a diverse set of videos, we adopt multiple candidate generators, as explained in the candidate generation section.

### Cold-start problem

How does our system handle the cold-start problem?

**For new users:** We don't have any interaction data about new users when they begin using our platform.

In this case, predictions are made using two-tower neural networks based on features such as age, gender, language, location, etc. The recommended videos are personalized to some extent, even for new users. As the user interacts with more videos, we are able to make better predictions based on new interactions.

**For new videos:** When a new video is added to the system, the video metadata and content are available, but no interactions are present. One way to handle this is to use heuristics. We can display videos to random users and collect interaction data. Once we gather enough interactions, we fine-tune the two-tower neural network using the new interactions.

### Training scalability

It's challenging to train models on large datasets in a cost-effective manner. In recommendation systems, new interactions are continuously added, and the models need to quickly adapt to make accurate recommendations. To quickly adapt to new data, the models should be able to be fine-tuned.

In our case, the models are based on neural networks and designed to be easily fine-tuned.

- The exploration-exploitation trade-off in recommendation systems [9].
- Different types of biases may be present in recommendation systems [10].
- Important considerations related to ethics when building recommendation systems [11].
- Consider the effect of seasonality - changes in users' behaviors during different seasons - in a recommendation system [12].
- Optimize the system for multiple objectives, instead of a single objective [13].
- How to benefit from negative feedback such as dislikes [14].
- Leverage the sequence of videos in a user's search history or watch history [2].

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